WORKED SOLUTION

STEP III 2009, Question 10

Mechanics • Simple Harmonic Motion

STEP III 2009, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/09-s3-q10

The key idea

The equilibrium compression d is there to tell us the stiffness of the spring without ever naming it: in equilibrium the spring force balances the weight, so $k/m = g/d$. After that, Newton's second law gives simple harmonic motion about the equilibrium position, and the drop height h only enters through the initial speed. The particle leaves the spring when the spring can no longer push on it, which happens when the compression returns to zero.

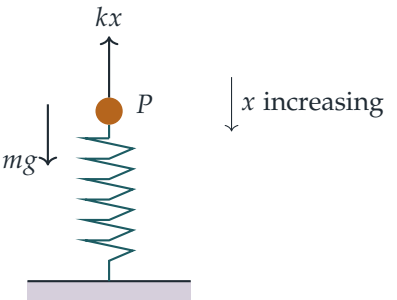
Solution

The differential equation

Model P as a particle of mass m and the spring as light, obeying Hooke's law with stiffness k, and ignore air resistance. When P rests on the spring, the forces on it are its weight mg down and the spring force kd up, so

$$kd = mg, \qquad \text{that is,} \qquad \frac{k}{m} = \frac{g}{d}.$$

Now take the dropped case. Measure x downwards (it is the compression), with $t = 0$ at the moment of impact. While P is in contact, the forces on it are as shown.



Because the spring is light, the force it exerts on P is exactly kx upwards (a light spring transmits the force at its ends). Newton's second law, taking downwards as positive, gives $m\ddot{x} = mg - kx$, so

$$\ddot{x} + \frac{g}{d}x = g, \qquad 0 \le t \le T.$$

Solving it

This is simple harmonic motion about $x = d$ with

$$\omega = \sqrt{g/d}.$$

A particular solution is the constant $x = d$, and the complementary function is $B\cos\omega t + C\sin\omega t$, so the general solution is

$$x = d + B\cos\omega t + C\sin\omega t.$$

The initial conditions come from the fall. At $t = 0$ the spring has just been touched, so $x(0) = 0$; and P has fallen freely through height h, so its speed is $\sqrt{2gh}$, downwards, which is the direction of increasing x. Hence

$$\begin{aligned} x(0) = d + B = 0 \qquad &\Rightarrow \quad B = -d, \\ \dot{x}(0) = C\omega = \sqrt{2gh} \qquad &\Rightarrow \quad C = \sqrt{2gh}\,\sqrt{d/g} = \sqrt{2hd}. \end{aligned}$$

So

$$x = d - d\cos\omega t + \sqrt{2hd}\,\sin\omega t, \qquad A = d, \quad B = -d, \quad C = \sqrt{2hd}, \quad \omega = \sqrt{g/d}.$$

The time of separation

Ask yourself When exactly does P lose contact?

The spring is not attached to P, so it can push but never pull. The contact force on P is the spring force kx, which stays positive while $x > 0$ and falls to zero when x returns to 0, with P moving upwards. If the motion carried on beyond that point, the equation would need a downward pull from the spring, which it can't give. So T is the first positive time with $x = 0$.

To find it, write the oscillating part as a single cosine. Let $R = \sqrt{d^2 + 2hd}$ and define the acute angle φ by

$$\cos\varphi = \frac{d}{R}, \qquad \sin\varphi = \frac{\sqrt{2hd}}{R},$$

so that $\tan\varphi = \sqrt{2h/d}$ and $\varphi = \arctan\sqrt{2h/d}$. Then $d\cos\omega t - \sqrt{2hd}\sin\omega t = R\cos(\omega t + \varphi)$, and

$$x = d - R\cos(\omega t + \varphi).$$

So $x = 0$ exactly when $\cos(\omega t + \varphi) = d/R = \cos\varphi$. As t increases from 0, the angle $\omega t + \varphi$ starts at φ, and on the interval $\varphi < \omega t + \varphi < 2\pi - \varphi$ we have $\cos(\omega t + \varphi) <$

$\cos\varphi$, so $x > 0$ there. The first return to $\cos\varphi$ is at

$$\omega T + \varphi = 2\pi - \varphi, \qquad \text{that is,} \qquad \omega T = 2\pi - 2\varphi.$$

Dividing by $\omega = \sqrt{g/d}$,

$$T = \sqrt{d/g}\left(2\pi - 2\arctan\sqrt{2h/d}\right),$$

as required.

Checks

- If $h = 0$ the particle is placed on the spring at its natural length with zero speed, $\varphi = 0$, and $T = 2\pi\sqrt{d/g}$: a full period, after which P is back at the top with zero speed. That is exactly right.
- As $h \to \infty$, $\varphi \to \frac{\pi}{2}$ and $\omega T \to \pi$: a very fast particle spends only half a period on the spring, because the equilibrium shift d becomes negligible next to the amplitude.
- By energy conservation, P should leave with speed $\sqrt{2gh}$. Indeed $\dot{x}(T) = R\omega\sin(2\pi - \varphi) = -R\omega\sin\varphi = -\sqrt{2gh}$.
- Numerically, stepping the motion with $d = 0.1\,\text{m}$, $h = 0.2\,\text{m}$ and $g = 9.8\,\text{m}\,\text{s}^{-2}$ gives a contact time of $0.411\,\text{s}$, matching the formula. Random values of d and h agree to six decimal places.

Common slips

- Measuring from the equilibrium position and then forgetting that x in the question is the compression, which gives $A = 0$.
- Taking the initial speed as $\sqrt{2gh}$ but with the wrong sign: downwards is the direction of increasing x.
- Assuming separation happens at the equilibrium position $x = d$, or at half a period. The spring keeps pushing until it is back to its natural length.

Beyond STEP

This is the simplest example of a *unilateral constraint*: the spring can push but not pull. The motion obeys different differential equations in different phases, which are stitched together by matching position and velocity, just as the free fall here is matched to the spring phase. The same idea runs through the theory of impacts and bouncing systems. For more on oscillators and on energy methods like the check above, see Kleppner and Kolenkow, *An Introduction to Mechanics*.

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